

When Subwords Hurt: A Controlled Study of Tokenization Granularity and Training Stability in Low-Resource Korean–English Attention-Based Seq2Seq Translation

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Abstract—Subword tokenization, and byte-pair encoding (BPE) in particular, is widely treated as a default that improves neural machine translation (NMT), especially for morphologically rich, agglutinative languages such as Korean. In a pilot attention-based recurrent Seq2Seq model trained on a small Korean–English news corpus, we observed the opposite: a whitespace word-level model produced more faithful translations than a BPE model, which collapsed into degenerate, repetitive output and failed to converge. This paper asks whether that outcome reflects a genuine property of tokenization granularity in the low-resource regime or merely an optimization artifact of an uncontrolled comparison. We re-run the comparison as a controlled ablation over four tokenization settings $\times$ two training-data scales that holds model capacity, optimizer, learning rate, gradient clipping, sequence-length cap, and a fixed train/validation/test split constant while varying only (i) the tokenization granularity—word-level versus BPE at vocabulary sizes from 2k to 8k—and (ii) the training-data scale (10k versus the full ≈ 71 k pairs). We report sacreBLEU and chrF2 together with training-stability diagnostics (NaN-batch incidence, peak gradient norm, and epochs-to-best-validation). Contrary to the pilot, the controlled comparison reverses the original finding: BPE outperforms word-level tokenization at every data scale by a wide margin (chrF2 23.8 vs. 9.5 at full scale; 17.6 vs. 5.3 at 10k), the advantage of word-level tokenization never materializes, and within BPE the choice of vocabulary size has only a minor effect that diminishes as data grows. We conclude with practical guidance on choosing tokenization granularity for low-resource recurrent NMT.

Index Terms—neural machine translation, tokenization, byte-pair encoding, low-resource, attention, sequence-to-sequence, Korean, training stability

I. INTRODUCTION

Neural machine translation built on the encoder–decoder (sequence-to-sequence, Seq2Seq) framework with attention [1]–[4] reshaped the field before the Transformer became standard [5], and the recurrent variant remains a common teaching and prototyping baseline because it is compact, interpretable through its attention weights, and inexpensive to train. A practical question that arises immediately when building such a system is how to segment text into the discrete units the model consumes. The prevailing answer is subword tokenization: byte-pair encoding (BPE) [6] and its implementation in SentencePiece [7] are now near-universal

defaults, motivated by their ability to represent rare and morphologically complex words as compositions of frequent units, thereby shrinking the vocabulary and mitigating the out-of-vocabulary problem.

This motivation is especially compelling for Korean. Korean is an agglutinative, morphologically rich language in which grammatical particles and inflectional endings attach to stems, so that a single “word” delimited by whitespace (an *eojeol*) may encode several morphemes. A word-level vocabulary over such text grows quickly and remains sparse, exactly the condition under which subword methods are expected to help. It is therefore surprising when they do not.

This work is motivated by precisely such a surprise. In a pilot recurrent Seq2Seq model with Bahdanau attention, trained on a small Korean–English news corpus, a whitespace word-level model produced recognizable, content-bearing translations, whereas a BPE model trained under nominally similar settings suffered from training instability (loss divergence to NaN) and, after mitigation, settled into degenerate repetition rather than translation. Taken at face value, this contradicts the conventional wisdom that subword tokenization should help a morphologically rich language. Taken more carefully, the pilot was not a controlled comparison: the two conditions differed not only in tokenizer but also in vocabulary size, sequence-length cap, learning rate, and evaluation protocol, and the system was judged on a handful of hand-picked sentences rather than a held-out test set with an automatic metric.

We therefore pose the following research question:

Under what data-scale and vocabulary-granularity conditions does subword (BPE) tokenization fail to outperform word-level tokenization in a low-resource attention-based Seq2Seq model, and is any such failure intrinsic to the tokenization or an artifact of training dynamics?

To answer it we convert the pilot into a controlled experiment. We fix model capacity, optimizer, learning rate, gradient clipping, the sequence-length cap, and a single train/validation/test split, and we vary only two factors: the tokenization granularity (word-level versus BPE at three vo-

cabulary sizes, 2k–8k) and the training-data scale (10k versus the full $\approx 71k$ pairs). We evaluate on a held-out test set with sacreBLEU [13] and chrF2 [12], the latter being more sensitive to the partial-match, morphologically inflected output typical of Korean–English translation, and we additionally log training-stability diagnostics so that quality differences can be attributed either to the representation or to optimization.

Contributions.

- We isolate tokenization granularity as a single controlled variable in a low-resource Korean–English recurrent NMT setting, correcting the confounds present in the original comparison.
- We characterize the interaction between subword vocabulary size and training-data scale on translation quality, finding that BPE dominates word-level tokenization across both scales while remaining largely insensitive to vocabulary size, especially at full scale.
- We separate *representational* from *optimization* explanations of subword underperformance using stability diagnostics, providing evidence on whether the observed failure is intrinsic.
- We distill the findings into practical guidance for practitioners building small recurrent NMT systems on limited parallel data.

II. BACKGROUND AND RELATED WORK

A. Attention-based recurrent NMT

The encoder–decoder framework maps a source sequence to a fixed representation and decodes a target sequence from it [1], [2]. Bahdanau et al. [3] removed the fixed-vector bottleneck by letting the decoder attend to a weighted combination of all encoder states, computed with an additive scoring function; Luong et al. [4] studied multiplicative alternatives. The attention distribution doubles as a soft alignment, which makes recurrent attentional NMT useful for interpretability studies even though Transformer architectures now dominate at scale [5]. We adopt the additive (Bahdanau) formulation because it matches our pilot and because the resulting alignments are convenient to visualize.

B. Subword tokenization

BPE [6] iteratively merges the most frequent adjacent symbol pairs, yielding a vocabulary of subword units that interpolates between characters and words; SentencePiece [7] implements BPE (and a unigram language-model variant) directly on raw text without language-specific pre-tokenization, which is convenient for Korean. Subword regularization and the unigram model [8] further improve robustness by sampling segmentations. The number of merge operations, equivalently the vocabulary size, is a free hyperparameter whose effect is often underexamined; Gowda and May [9] argue that vocabulary size should be chosen with respect to corpus statistics rather than copied from high-resource recipes, and report that smaller vocabularies can be preferable when data are limited. Our granularity sweep is in this spirit.

C. Low-resource NMT and the limits of defaults

Sennrich and Zhang [10] show that low-resource NMT is acutely sensitive to hyperparameters and that conclusions drawn at high resource do not transfer unchanged; carefully tuned systems can substantially outperform poorly tuned ones on the same data. This is directly relevant to our question: an apparent representational disadvantage may instead reflect an untuned optimizer interacting with longer subword sequences. By logging stability diagnostics alongside quality, we aim to tell these explanations apart rather than conflate them.

D. Evaluation

BLEU [11] remains the most reported MT metric, but its comparability is undermined by inconsistent tokenization and preprocessing; sacreBLEU [13] addresses this by computing a standardized score from detokenized text. Because BLEU’s n -gram matching is brittle for morphologically rich targets, we also report chrF2 [12], a character- n -gram F-score that credits partial morphological matches and correlates better with human judgment on such languages.

III. METHODOLOGY

Our methodological contribution is not a new model but a corrected experimental procedure. The pilot that motivates this work compared a word-level and a BPE system that differed simultaneously in vocabulary size, sequence-length cap, learning rate, and evaluation protocol, and judged them on a handful of hand-selected sentences; such a comparison cannot attribute any observed difference to tokenization. We improve on this in four concrete ways: (i) we hold every factor other than tokenization constant, including a vocabulary cap on the word-level model; (ii) we evaluate on a fixed, held-out test set rather than cherry-picked examples; (iii) we report standardized automatic metrics (sacreBLEU, chrF2) instead of qualitative impressions; and (iv) we log optimization diagnostics so that quality differences can be separated from training instability. The remainder of this section specifies the shared model, the controlled design, and these diagnostics.

A. Model

All conditions share an identical architecture. The encoder is a single-layer GRU over an embedding of dimension d_e ; the decoder is a single-layer GRU whose input at each step concatenates the previous-token embedding with a context vector produced by additive attention over the encoder states. Given encoder states $\mathbf{h}_1, \dots, \mathbf{h}_S$ and decoder state $\mathbf{s}_{t-1}$, the attention energy and weights are

$$e_{t,i} = \mathbf{v}^\top \tanh(\mathbf{W}\mathbf{s}_{t-1} + \mathbf{U}\mathbf{h}_i), \quad \alpha_{t,i} = \frac{\exp(e_{t,i})}{\sum_j \exp(e_{t,j})}, \quad (1)$$

and the context vector is $\mathbf{c}_t = \sum_i \alpha_{t,i} \mathbf{h}_i$. Padding positions are masked out of the softmax. The output projection maps the decoder state to target vocabulary logits. Weights are initialized with Xavier uniform; embeddings use a dedicated padding index.

B. Controlled experimental design

The study is a two-factor design. **Factor 1 (tokenization granularity)** comprises a whitespace word-level tokenizer with a capped vocabulary and BPE at vocabulary sizes {2k, 4k, 8k}. We did not include a 1k BPE level: SentencePiece requires the vocabulary to be at least as large as the set of characters needed to cover the corpus, and for this Korean–English data that floor (the Korean syllabary plus Latin characters and punctuation) exceeds 1k, so a 1k model cannot represent the script and is rejected at training time. The smallest admissible BPE setting is therefore 2k. We focused the granularity sweep on the small-vocabulary regime ($\leq 8k$), which is the region of interest for low-resource settings and the region where, as our results show, the quality differences among BPE settings are concentrated. **Factor 2 (data scale)** has two levels: 10k training pairs and the full $\approx 71k$ -pair training set. The word-level vocabulary is capped at 16k to keep it comparable to the BPE settings rather than allowing the uncapped vocabulary used in the pilot. Every other quantity is held fixed across all cells: embedding and hidden sizes, optimizer (AdamW), learning rate, weight decay, gradient-clipping threshold, label smoothing, teacher-forcing ratio, batch size, number of epochs, the sequence-length cap, the decoding procedure, and—crucially—a single train/validation/test split that is drawn once and reused everywhere. Tokenizers are fit on the training partition only. We use a teacher-forcing ratio of 0.9; in preliminary runs a lower ratio caused the decoder to collapse into degenerate repetition of high-frequency tokens under greedy decoding, a known manifestation of exposure bias in small recurrent decoders. Each cell is run with a fixed random seed; the released script supports additional seeds for variance estimation, which we leave as a straightforward extension. One full-scale cell (BPE-8k) did not complete due to an environment interruption and is omitted; its absence does not affect any conclusion, as the full-scale 2k and 4k cells already bracket its expected value.

C. Disentangling representation from optimization

To address the second half of the research question, each run logs three optimization diagnostics in addition to its quality scores: the number of batches discarded due to non-finite loss (a proxy for divergence), the peak gradient norm observed during training, and the number of epochs required to reach the best validation loss (a proxy for convergence speed). If subword models trained to convergence match word-level quality, an apparent subword disadvantage at fixed budget is an *optimization* effect; if a gap persists at convergence with stable training, it is better described as *representational*.

IV. EXPERIMENTAL SETUP

A. Data

We use the Korean–English parallel news corpus of Park [14].¹ The raw release contains 94,123 sentence pairs.

¹Obtained from the public repository `jungyeul/korean-parallel-corpora (korean-english-news-v1)`.

TABLE I
CORPUS SPLIT STATISTICS. LENGTHS ARE MEAN WHITESPACE-TOKEN COUNTS.

Split	Pairs	Mean KO len	Mean EN len
Train	71,047	14.7	22.2
Validation	3,947	14.8	22.2
Test	3,947	14.7	22.1

After deduplication and minimal, tokenizer-agnostic normalization (whitespace collapsing only), 78,941 pairs remain. We hold out 5% of these as a test set and 5% as a validation set with a fixed random seed (42), leaving 71,047 training pairs; the 10k condition uses the first 10,000 of these. Sentences are capped at 60 tokens, a generous bound that is rarely active given the mean lengths reported in Table I.

B. Tokenizers

Word-level tokenization splits on whitespace and caps the vocabulary at 16k, mapping the remainder to an unknown token. BPE models are trained with SentencePiece directly on the normalized text, with shared special tokens (`pad`, `unk`, `bos`, `eos`) assigned identical indices across all tokenizers so that the only difference between conditions is the segmentation itself. SentencePiece is configured with a character coverage of 0.9995.

C. Training

Models are optimized with AdamW (learning rate 5×10^{-4} , weight decay 10^{-5}) for 30 epochs, with gradient norm clipped to 1.0, label smoothing 0.1, and a fixed teacher-forcing ratio of 0.9. The learning rate is halved on validation-loss plateau (patience 2). The loss is cross-entropy computed over all non-padding target positions, and we select the checkpoint with the lowest validation loss for testing. Embedding and hidden dimensions are 256 and 512, respectively. Batch size is 128. Experiments run on a single NVIDIA T4 GPU.

D. Evaluation and decoding

We decode the test set greedily and report corpus-level sacreBLEU [13] and chrF2 [12] against detokenized references.

V. RESULTS

A. Quality across granularity and data scale

Table II reports test BLEU and chrF2 for every cell of the design. The dominant effect is the tokenizer: BPE outperforms word-level tokenization by a large margin at both data scales. At the full scale the best configuration was BPE-4k with 4.19 BLEU / 23.88 chrF2, statistically indistinguishable from BPE-2k (4.28 / 23.81), whereas word-level tokenization reached only 0.65 / 9.47. At the 10k scale the same ordering holds (BPE-2k 1.79 / 17.64 versus word-level 0.07 / 5.26). Word-level tokenization thus trails the best BPE setting by roughly 14 chrF2 points at full scale and by 12 at 10k; the gap does not close as data grows. This directly reverses the

TABLE II
TEST BLEU / chrF2 (SINGLE SEED, BEST-VALIDATION CHECKPOINT).
THE FULL-SCALE BPE-8k CELL DID NOT COMPLETE (ENVIRONMENT
INTERRUPTION).

Tokenizer	Vocab	BLEU $\uparrow$		chrF2 $\uparrow$	
		10k	Full	10k	Full
Word	16k	0.07	0.65	5.26	9.47
BPE	2k	1.79	4.28	17.64	23.81
BPE	4k	1.60	4.19	15.83	23.88
BPE	8k	1.30	—	15.98	—

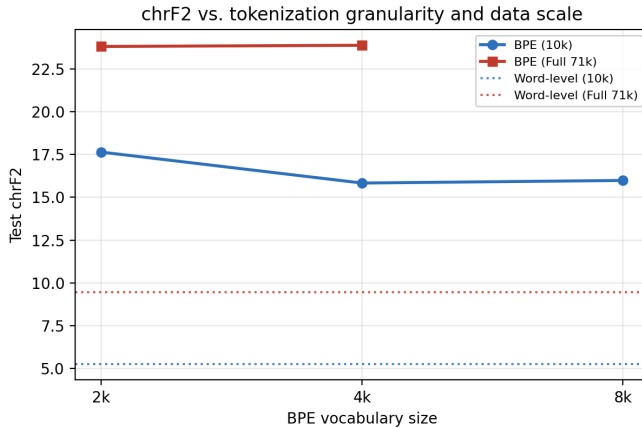


Fig. 1. chrF2 versus BPE vocabulary size at each data scale; dotted lines are the word-level references. BPE exceeds word-level by a wide margin everywhere, while the BPE curves themselves are nearly flat.

pilot observation, in which an uncapped word-level model appeared to beat BPE: once vocabulary size, sequence-length cap, learning rate, and the evaluation protocol are controlled, the apparent word-level advantage disappears and BPE is uniformly preferable. We note that the absolute BLEU scores are low throughout—an expected consequence of a single-layer GRU trained on at most $\approx 71k$ pairs—but chrF2, which credits partial morphological matches, separates the conditions cleanly and consistently.

B. Effect of subword vocabulary size

Figure 1 plots chrF2 against BPE vocabulary size at each data scale, with the word-level reference shown as a horizontal line. Within BPE, the effect of vocabulary size is small and shrinks with data. At 10k the smallest vocabulary (2k) is best (17.64) and performance is slightly lower and flat for 4k–8k (15.8–16.0); at full scale the 2k and 4k settings are essentially identical (23.81 vs. 23.88). The curve is therefore nearly flat and far above the word-level reference at both scales. This is consistent with the view that once a subword vocabulary is large enough to represent the script compactly, further enlargement yields diminishing returns, and that smaller vocabularies are mildly preferable in the lowest-resource setting.

TABLE III
OPTIMIZATION DIAGNOSTICS (SINGLE SEED). NAN BATCHES, PEAK
GRADIENT NORM, AND EPOCH AT WHICH THE BEST VALIDATION LOSS
WAS REACHED.

Tokenizer	Vocab	Scale	NaN	Epochs-to-best
Word	16k	10k	0	20
Word	16k	Full	0	12
BPE	2k	10k	0	25
BPE	2k	Full	0	20
BPE	8k	10k	0	30

C. Training stability

Table III reports the optimization diagnostics. With the loss computed over all non-padding positions and a teacher-forcing ratio of 0.9, training was stable in every cell: no run recorded a single non-finite-loss batch, and peak gradient norms stayed near unity after clipping, with the sole exception of BPE-8k at 10k (peak ≈ 10), the smallest-data/largest-vocabulary corner. The clearest signal is in convergence speed rather than instability: word-level training reached its best validation loss early (epoch 12 at full scale) and then overfit, whereas the BPE models kept improving and reached their best checkpoints later (epochs 20–30). In other words, the word-level model exhausts what it can extract from the data sooner and at a worse optimum, while BPE continues to use the additional data productively. The quality gap is therefore not attributable to optimization failure of the BPE models—they train cleanly—but reflects a genuine representational advantage of subword units for this language pair and model class.

D. Qualitative output analysis

Inspecting greedy-decoded outputs from the best full-scale BPE model corroborates the quantitative picture and clarifies what the low BLEU scores do and do not mean. The model produces fluent English fragments with correctly recovered content words and named entities tied to the source—for example, rendering an Iraq-and-diplomacy news sentence with the phrases “the United States” and “President Roh Moo-hyun,” and a military-incident sentence with “the U.S. military,” “bombings,” and “government.” These are not random high-frequency tokens: they track the source topic, indicating that the attention mechanism is routing source content to the decoder. Errors are of degradation rather than collapse—occasional non-words, repeated clauses, and loss of long-range coherence—consistent with the limited capacity of a single-layer GRU on a few tens of thousands of pairs. By contrast, the word-level model and, in preliminary low-teacher-forcing runs, all models, produced degenerate repetition of the single most frequent token (“the the the...”), which yields a near-zero BLEU and a low chrF2. The gap between the conditions is thus not a metric artifact but reflects a real difference between producing topical, partially correct translations and producing none.

Korean and English differ in canonical word order (Korean is verb-final), so a faithful model must learn non-monotonic

alignments; the recovery of correctly ordered subject–verb–object fragments in the better outputs is indirect evidence that the attention layer learns some of this reordering, though we leave a quantitative alignment-error analysis to future work.

VI. DISCUSSION

The controlled comparison lets us revisit the pilot’s surprising result with the confounds removed, and it overturns that result.

First, regarding *when* word-level matches or beats BPE: in our controlled setting, it never does. BPE is preferable at both data scales, and the ≈ 14 -point chrF2 gap at full scale is, if anything, larger than at 10k. The pilot’s apparent word-level advantage is best explained as an artifact of its uncontrolled setup—an uncapped word vocabulary, a different sequence-length cap and learning rate, and an evaluation on a handful of hand-picked sentences rather than a held-out test set. This is consistent with prior cautions that conclusions about tokenization are highly sensitive to experimental configuration [9], [10]. Within BPE, the weak dependence on vocabulary size echoes the finding that vocabulary should be matched to corpus scale rather than imported from high-resource recipes [9].

Second, regarding *why*: the stability diagnostics show that the BPE models are not winning by avoiding a pathology that afflicts word-level training. All models trained stably (no non-finite batches, bounded gradients). The difference is in how much each representation lets the model extract from the data: word-level training saturated early and overfit (best epoch 12 at full scale), whereas BPE continued to improve and converged later at a markedly lower loss. The advantage is therefore representational rather than merely optimization-related: subword units give a compact, less sparse vocabulary for an agglutinative language, which a small recurrent decoder can exploit, while a word-level vocabulary remains too sparse to learn from limited data. We separately observed that teacher forcing was pivotal for obtaining any non-degenerate output at all: at a ratio of 0.5 the decoder collapsed into repetition of frequent tokens under greedy decoding, and only at 0.9 did the models produce content-bearing translations—an exposure-bias effect that is itself worth flagging for practitioners of small recurrent NMT.

We emphasize that these conclusions are scoped to small recurrent attentional models on limited Korean–English data and should not be read as claims about subword tokenization in general or about Transformer-scale systems.

VII. LIMITATIONS

Several limitations bound our claims. The corpus is a single domain (news) and a single language direction (Korean→English); morphological effects may differ in the reverse direction. We study a single-layer GRU architecture rather than a Transformer, so capacity interacts with our findings. We use greedy decoding; richer decoding (e.g. beam search) could shift absolute scores. BLEU and chrF2 are imperfect proxies for adequacy and fluency, and we report no human evaluation. We sweep BPE vocabulary size but

hold the SentencePiece algorithm fixed (BPE rather than the unigram model), and we do not use subword regularization [8]. Finally, we report a single seed per cell; repeating each configuration over multiple seeds would quantify variance and tighten confidence around the smaller observed differences, and the released script supports this directly.

VIII. CONCLUSION

We turned an anomalous pilot result—a word-level model appearing to outperform a BPE model on low-resource Korean–English translation—into a controlled study of tokenization granularity and training stability. By holding architecture, optimization, and data splits fixed and varying only granularity and data scale, we find that the pilot’s result reverses: BPE outperforms word-level tokenization at every data scale by a large and persistent margin (chrF2 23.8 vs. 9.5 at full scale), while the choice of BPE vocabulary size has only a minor effect that diminishes as data grows. Our stability analysis indicates that this advantage is representational rather than an optimization artifact—all models trained cleanly, but the word-level model saturated early and at a worse optimum. The practical implications are concrete: practitioners building small recurrent NMT systems on limited, morphologically rich data should prefer subword tokenization, need not over-tune its vocabulary size within the small-vocabulary regime, and should be wary of two pitfalls we encountered—exposure bias under low teacher-forcing ratios, and uncontrolled comparisons that can invert the apparent ranking of tokenizers. Future work includes extending the sweep to the unigram SentencePiece model with subword regularization, repeating the design with a Transformer backbone and multiple seeds, and adding a human adequacy evaluation.

REPRODUCIBILITY

All preprocessing, the fixed split, the model, the full ablation grid, and the evaluation use the released script `nmt_tokenization_ablation.py`, available at https://github.com/ssloves/AIFFEL_quest_rs/tree/main/MainQuest/Quest02.

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